WORKED SOLUTION

# STEP II 2025, Question 3

Pure • Comparing Powers

**STEP II 2025, Question 3**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Every comparison of the form $a^b$ against $b^a$ (with $a, b > 0$) becomes, after taking logarithms and dividing by $ab$, a comparison of $\frac{\ln a}{a}$ with $\frac{\ln b}{b}$. So everything comes down to the graph of $f(x) = \frac{\ln x}{x}$: it rises to a single peak at $x = \mathrm{e}$ and then falls. Two numbers on the same side of e are easy to compare; two on opposite sides need a stepping stone, such as the neat coincidence $f(2) = f(4)$.

## Solution

Throughout, write $f(x) = \dfrac{\ln x}{x}$ for $x > 0$. Since ln is increasing, for positive $a, b$

$$a^b > b^a \iff b \ln a > a \ln b \iff \frac{\ln a}{a} > \frac{\ln b}{b} \iff f(a) > f(b),$$

where we divided by $ab > 0$. We use this translation in every part.

### Part (i)

By the quotient rule,

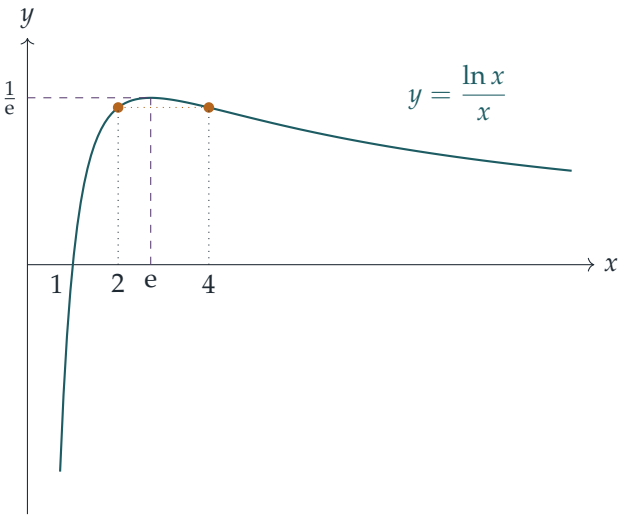
$$f'(x) = \frac{\frac{1}{x} \cdot x - \ln x}{x^2} = \frac{1 - \ln x}{x^2},$$

which is positive for $0 < x < \mathrm{e}$ and negative for $x > \mathrm{e}$. So $f$ is strictly increasing on $(0, \mathrm{e}]$ and strictly decreasing on $[\mathrm{e}, \infty)$, with a maximum value $f(\mathrm{e}) = 1/\mathrm{e}$.

Other features:

- $f(1) = 0$; $f < 0$ on $(0, 1)$ and $f > 0$ on $(1, \infty)$.
- As $x \to 0^+$, $\ln x \to -\infty$ while $1/x \to +\infty$, so $f(x) \to -\infty$: the $y$-axis is an asymptote.
- As $x \to \infty$, $f(x) \to 0^+$ because $\ln x$ grows more slowly than $x$: the $x$-axis is an asymptote.

- Differentiating again gives $f''(x) = \dfrac{2\ln x - 3}{x^3}$, so there is a point of inflection at $x = \mathrm{e}^{3/2} \approx 4.48$.




The amber points mark $f(2) = \frac{\ln 2}{2} = \frac{2\ln 2}{4} = f(4)$, which we shall use in part (iii).

**Part (ii)**

**(a)** By the translation, $3^\pi > \pi^3 \iff f(3) > f(\pi)$. Both 3 and $\pi$ exceed e, and $f$ is strictly decreasing on $[\mathrm{e}, \infty)$; since $3 < \pi$ we get $f(3) > f(\pi)$, so $3^\pi > \pi^3$.

**(b)** Here the inequality is $\left(\frac{9}{4}\right)^{\sqrt{5}} > \sqrt{5}^{9/4}$, which is equivalent to $f\left(\frac{9}{4}\right) > f(\sqrt{5})$. Now both numbers lie below e: indeed $\sqrt{5} < \frac{9}{4} = 2.25 < \mathrm{e}$, where $\sqrt{5} < \frac{9}{4}$ because $5 < \frac{81}{16}$. On $(0, \mathrm{e}]$ the function $f$ is strictly increasing, so the larger input gives the larger value: $f\left(\frac{9}{4}\right) > f(\sqrt{5})$, as required.

**Part (iii)**

With $a = x$ and $b = x + 2$, the translation gives

$$x^{x+2} > (x+2)^x \iff f(x) > f(x+2).$$

For $1 < x < 2$ we have $x \in (1, 2)$ and $x + 2 \in (3, 4)$, on opposite sides of e, so we need a stepping stone.

> **?** **Ask yourself** How can we compare $f$ at a point left of e with $f$ at a point right of it?

Use the horizontal line through the two amber points. Since $1 < x < 2 < \mathrm{e}$ and $f$ is increasing there, $f(x) < f(2)$. Since $\mathrm{e} < x + 2 < 4$ and $f$ is decreasing there,

$f(x+2) > f(4)$. And $f(2) = f(4)$, so

$$f(x) < f(2) = f(4) < f(x+2).$$

Hence $f(x) < f(x+2)$, and so $(x+2)^x$ is the larger: $(x+2)^x > x^{x+2}$ for every $x$ in $(1,2)$.

**Part (iv)**

Rewrite both sides of the first inequality as powers of 3 and of $2\sqrt{2}$:

$$9^{\sqrt{2}} = \left(3^2\right)^{\sqrt{2}} = 3^{2\sqrt{2}}, \qquad \sqrt{2}^9 = \left(\sqrt{2}^3\right)^3 = \left(2\sqrt{2}\right)^3.$$

So the two inequalities have identical left-hand sides and identical right-hand sides; they are the same statement and hence equivalent.

By the translation, $3^{2\sqrt{2}} > (2\sqrt{2})^3 \iff f(3) > f(2\sqrt{2})$. Now $\mathrm{e}^2 < 8$ gives $\mathrm{e} < \sqrt{8} = 2\sqrt{2}$ (taking positive square roots), and $2\sqrt{2} < 3$ since $8 < 9$. So

$$\mathrm{e} < 2\sqrt{2} < 3,$$

and because $f$ is strictly decreasing on $[\mathrm{e}, \infty)$, $f(2\sqrt{2}) > f(3)$. Therefore $(2\sqrt{2})^3 > 3^{2\sqrt{2}}$, that is,

$$\sqrt{2}^9 > 9^{\sqrt{2}}: \quad \text{the larger is } \sqrt{2}^9.$$

**Part (v)**

Let $a = \sqrt[3]{3}$. We want to compare $8^a$ with $a^8$, that is, $f(8)$ with $f(a)$. Since $a < 2 < \mathrm{e} < 8$, the two inputs are again on opposite sides of the peak, and they turn out to be close ($f(a) \approx 0.254$, $f(8) \approx 0.260$), so we need something sharper than part (iii)'s trick.

? **Ask yourself** Can we write $f(a)$ and $f(8)$ in terms of $f$ at simpler points?

The useful identity is $f(x^k) = \dfrac{k \ln x}{x^k}$. It gives

$$f(a) = \frac{\frac{1}{3}\ln 3}{a} = \frac{f(3)}{a}, \qquad f(8) = \frac{3\ln 2}{8} = \frac{3}{4}f(2), \qquad f(2\sqrt{2}) = \frac{\frac{3}{2}\ln 2}{2\sqrt{2}} = \frac{3}{2\sqrt{2}}f(2).$$

From part (iv), $f(3) < f(2\sqrt{2})$. Dividing by $a > 0$,

$$f(a) = \frac{f(3)}{a} < \frac{f(2\sqrt{2})}{a} = \frac{3f(2)}{2\sqrt{2}\,a}.$$

Finally, $a = 3^{1/3} > 2^{1/2} = \sqrt{2}$, because raising both to the sixth power gives $9 > 8$. Hence $2\sqrt{2}\,a > 2\sqrt{2} \cdot \sqrt{2} = 4$, and since $f(2) > 0$,

$$f(a) < \frac{3f(2)}{2\sqrt{2}\,a} < \frac{3f(2)}{4} = f(8).$$

So $f(8) > f(a)$, which by the translation means $8^a > a^8$: the larger is $8^{\sqrt[3]{3}}$.

## Checks

Numerically, $3^{\pi} \approx 31.54$ against $\pi^3 \approx 31.01$; $\left(\frac{9}{4}\right)^{\sqrt{5}} \approx 6.131$ against $\sqrt{5}^{9/4} \approx 6.114$; at $x = 1.5$, $1.5^{3.5} \approx 4.13$ against $3.5^{1.5} \approx 6.55$; $9^{\sqrt{2}} \approx 22.36$ against $\sqrt{2}^{9} \approx 22.63$; and $8^{\sqrt[3]{3}} \approx 20.07$ against $\sqrt[3]{3}^{8} \approx 18.72$. All agree with the conclusions above. The margins are small in (ii) and (iv), which is exactly why the graph argument is needed rather than rough estimation. A computer check of part (iii) at $10^5$ random points in $(1,2)$ found no exceptions.

## Common slips

- Comparing $f$ at two points on opposite sides of e as if $f$ were monotonic. In (iii) and (v) a stepping stone is essential.
- Forgetting that dividing $b\ln a > a\ln b$ by $ab$ keeps the direction only because $ab > 0$.
- In (ii)(b), not justifying which of $\frac{9}{4}$ and $\sqrt{5}$ is larger, or that both lie below e.

## Beyond STEP

The equation $a^b = b^a$ with $a \neq b$ is exactly $f(a) = f(b)$, and every horizontal line $y = c$ with $0 < c < 1/\mathrm{e}$ cuts the graph twice. Its solutions with $a < b$ can be parametrised as $a = (1+\frac{1}{t})^t$, $b = (1+\frac{1}{t})^{t+1}$ for $t > 0$; taking $t = 1$ gives $(2,4)$, the only such pair of positive integers. As $t \to \infty$ both tend to e, one of the standard limits of first-year analysis. For the careful treatment of monotonicity and the mean value theorem behind arguments like these, see Abbott, *Understanding Analysis*.

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